

Take-Home Assessment

Time budget: 3–4 hours. Not more. We care about your reasoning, not polish. Use AI freely — we expect it and we'll ask you about it.

Where this fits

This take-home is one step in a short process: a quick technical chat, then this exercise, then a final interview. In that final interview you'll walk us through what you built, we'll dig into your reasoning together, and then we'll hand you a new requirement and ask you to implement it live. So build something you'll be comfortable defending and *extending* — not a one-off you'd be afraid to touch. We won't grade the artifact in isolation; the goal is for it to be *yours enough to defend*.

What we want built

A small data product. In plain terms:

- Ingest a parcels dataset and a zoning dataset, normalize them, and load them into a database.
- On top of that, be able to answer questions like *"show me the residential parcels bigger than an acre,"* and roll the results up into simple stats by area (e.g. count and median lot size per neighborhood).

Two specifics so we're all judging the same thing:

- **"Residential"** means any zoning district whose code or name begins with "R" or the word "Residential" (e.g. R-1, RM, "Residential — Single Family"). Use that rule.
- **Lot size** must be **computed from the parcel geometry** — do not read it from any stated area field in the source, because those are unreliable. Calculate the area yourself and convert it to acres.

It's still loosely specified in places, and that's on purpose — we haven't pinned everything down (for example, what "by area" should group on). Don't paper over gaps like that: make a call you can defend, and tell us what you'd go ask to resolve it.

The data

Use these two public datasets:

- **Zoning (City of Buda):** <https://services6.arcgis.com/vXZW4vAaPRr14z2s/ArcGIS/rest/services/Zoning/FeatureServer/0>
- **Parcels (Hays County):** <https://hays-county-haysgis.hub.arcgis.com/>

These are real, so they're messy. The two come from different sources and jurisdictions — the zoning covers a city, the parcels cover the whole county — and you'll find inconsistent codes, mixed-use classifications, missing or stale lot areas, and projection/units quirks. That's the point: we'd rather see how you handle the genuine article than a sandbox we cleaned for you. Spend a few minutes sizing up the data before you commit to an approach, and tell us in a line or two what you found wrong with it.

What to deliver

1. A short design document (a few pages, not a thesis):

- *System decomposition* — the main components/modules and their responsibilities.
- *A high-level architecture sketch* — one diagram is plenty.
- *Key decisions, one-line rationale each* — and, just as importantly, the decisions you **deferred** or deliberately chose not to make.

2. A requirements section. List the assumptions you made, and the questions you'd take back to stakeholders to resolve what the brief left unclear. This is where we look at how you'd elicit requirements: *what would you go ask, and why does it matter?* If you spotted a place where the requirements are ambiguous or pull in two directions, this is where to call it out.

3. A minimal working skeleton — enough to show the core is real, not just boxes on a diagram. Stub whatever isn't core. Concretely, the skeleton must include:

- The actual **data model** for parcels and zoning (your schema, with the normalization choices you made).
- A **sturdy ingestion pipeline** for the feeds. Nothing over-engineered — but not so fragile that running it twice, or against a corrected version of the data, leaves you with a mess.
- A way to answer the **core question**: find the residential parcels bigger than an acre — with lot size computed from the geometry — and roll the results up into simple stats by area (e.g. count and median lot size). How you structure that is your call.

Optional stretch (only if you have time — don't blow the budget): using PostGIS, an "average lot size within 1 km of a given point" query.

4. A brief note on your process. How you worked, how you used AI, and what checks you applied to its output.

What we're looking for

Not a finished product. We're looking at how you decompose a loosely-specified problem, where you draw the line on scope in four hours, which decisions you make versus consciously defer, and whether you ask the right questions instead of guessing. The data will force real judgment calls — relating city zoning to county-wide parcels, computing area correctly across projections and units, what to do with nulls and duplicates, and what to group the stats on. We want to see those calls made deliberately and explained, not hidden.

When something in your submission is thin, that's fine — we'll know what to talk about together in the final interview.